

TARGET ASSESSMENT · PSORIASIS

A small-molecule route to the **IL-23–IL- 23R axis** in psoriasis

Structural assessment, five evidence-based discovery strategies, and a preclinical target product profile (TPP) built from the public record.



TARGET

IL-23R–IL-23 (p19)

INDICATION

Moderate-to-severe plaque psoriasis

SCOPE

Preclinical TPP

BASIS

Public structures, trials, PK, safety

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A validated IL-23 target, a structural white space, and a small-molecule path

PATIENTS

~125M

people with psoriasis worldwide (JAMA 2020); US adult prevalence 3.0%

STRUCTURAL WHITE SPACE

0

PDB (Protein Data Bank) structures of small molecules bound to IL-23/IL-23R

INTERFACE

≈900 Å²

p19–IL-23R buried surface over 51 contact residues (Bloch 2018)

TPP TARGET

≥80%

week-16 sPGA 0/1; sPGA/IGA = global skin ratings, 0 = clear

VALIDATION

The target works; delivery is the gap

Three approved monoclonal antibodies (mAbs) clear 80–85% of patients by week 16, but all inject subcutaneously (SC) with 15–28 d half-lives. Oral small molecules (TYK2/PDE4) trail at 18–54%.

WHITE SPACE

No small molecule has bound this interface

The closest direct oral agent is icotrokinra, a macrocyclic peptide in phase 3 with week-16 IGA 0/1 of 68–70%.

THE PATH

Structure points to five entry points

Five discovery strategies ranked by evidence, and a preclinical TPP written only from verified public data.

IL-23 sits at the Th17 check-point that drives the psoriatic skin lesion

1

IL-23 binds the receptor

IL-23 (p19 + p40) binds IL-23R through its immunoglobulin D1 domain.

2

The β 1 subunit joins

IL-12R β 1, shared with the IL-12 receptor, is recruited in a ~100 $\times$ cooperative step.

3

Intracellular signaling

JAK1, JAK2 and TYK2 (intracellular kinases) activate STAT3 and STAT4 (signal transducers).

4

Th17 expansion

Th17 (IL-17-producing T-helper) cells expand and commit under IL-23.

5

Cytokine burst

IL-17A/F, IL-22, GM-CSF and CCL20 drive keratinocyte inflammation and the lesion.

DEPTH OF BLOCKADE MATTERS

Responders need >90% suppression of the IL-17 response (Wu 2024): partial blockade is not partial efficacy.

BIOMARKER CANDIDATE

ADAM17 shedding releases soluble IL-23R into serum (Franke 2016), a trackable pharmacodynamic readout.

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Psoriasis is common, polygenic, and genetically validated at IL-23R itself

WORLDWIDE

~125M

people affected (Armstrong & Read, JAMA 2020)

UNITED STATES

3.0%

adult prevalence, ≈7.55M people (NHANES 2011–14)

PSA CO-OCCURRENCE

19.7%

psoriatic arthritis co-occurs with psoriasis

CARDIOVASCULAR RISK

OR 1.27

odds ratio vs no psoriasis

GENETIC VALIDATION

- HLA-C*06:02 is the principal risk allele and a response biomarker.
- PSORS1 (CDSN) and 63 mapped loci; heritability ~28% (Tsoi 2017).
- IL23R R381Q protects against psoriasis and inflammatory bowel disease (IBD) (Duerr 2006; Capon 2007).
- A protective human mutation at the receptor is the strongest genetic case that blocking this axis is tolerable.

Efficacy moved down the pathway to the lesion, and a direct oral option just arrived

1

Anti-TNF mAbs

2008 onward. First approved psoriasis class, injected.

2

Anti-p40 mAbs

Ustekinumab 2012; blocks p40, shared with IL-12.

3

Anti-IL-17 mAbs

2015–16. Block the main effector cytokine in skin.

4

IL-23/IL-23R mAbs

Guselkumab 2017, tildrakizumab 2018, risankizumab 2019: the 80–85% week-16 bar.

5

Oral JAK/TYK2

Tofacitinib 2017, deucravacitinib 2022; convenience, class-safety flags, lower efficacy.

6

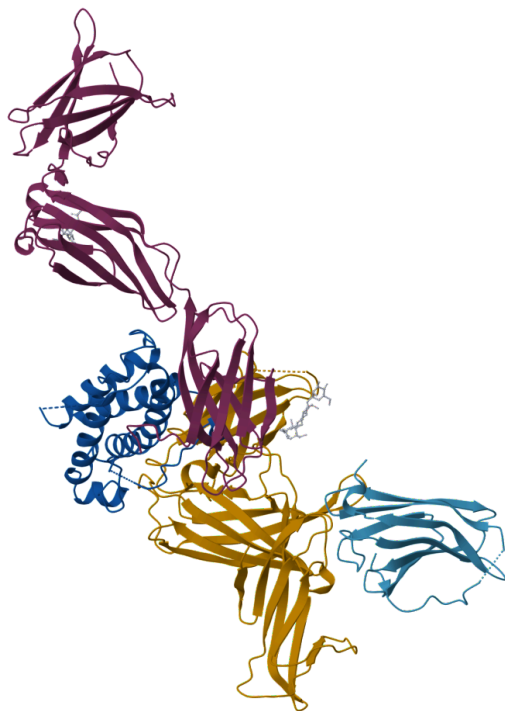
Oral peptide IL-23R antagonist

Icotrokinra: phase 3 results 2025. First direct oral agent on the axis; a macrocyclic peptide.

ONE ORAL PROTEIN PRECEDENT

The only other oral agent on this pathway in the literature is a VHH (variable heavy-heavy domain) nanobody, a single-domain antibody with a binding affinity (KD) of 128 pM (PNAS 2025). Oral protein precedent, still not a small molecule.

The complex is a two-tier interface: p40 anchors the receptor, p19 provides the specificity

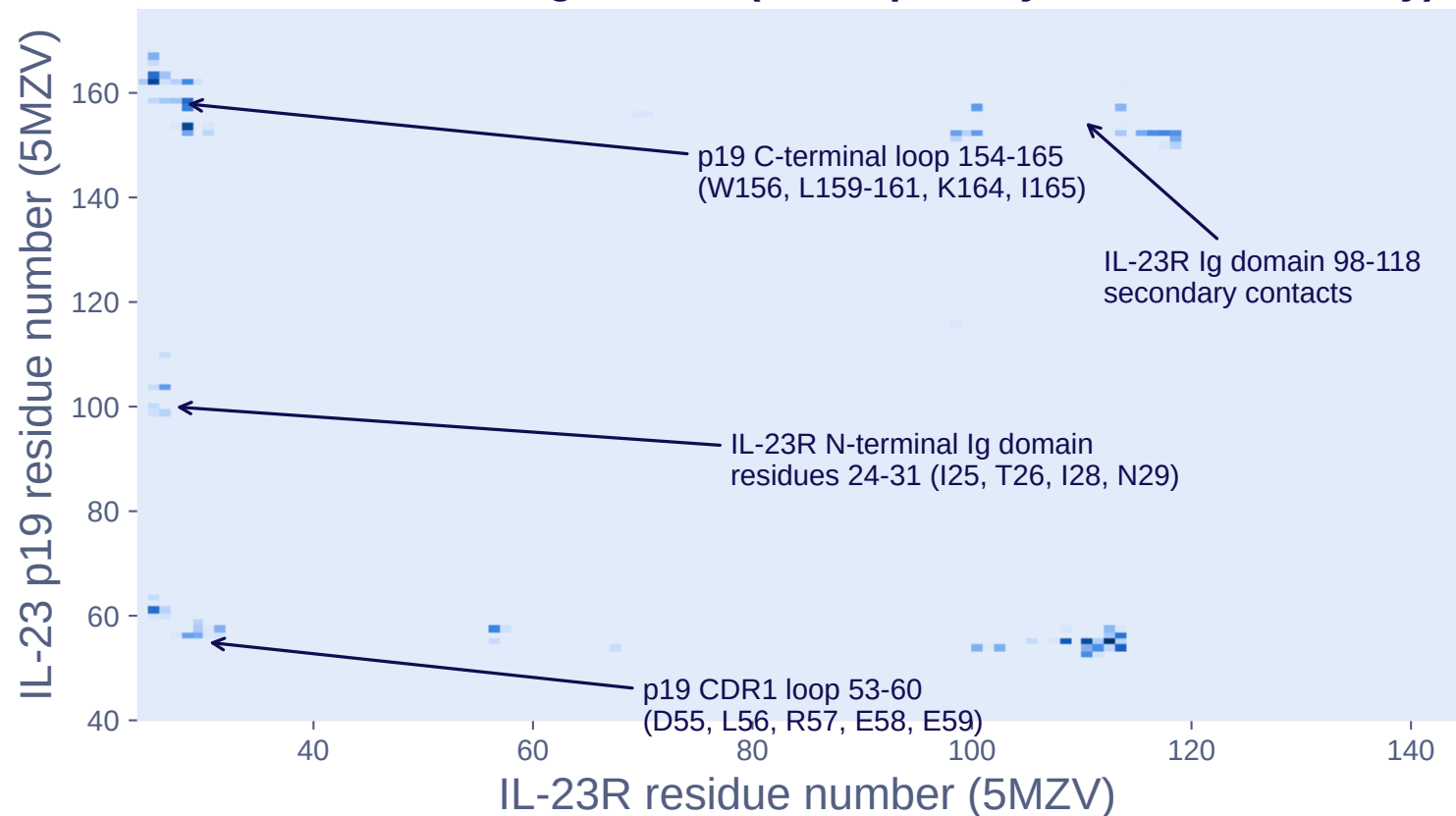


- p40 is the hub: 1249 contacts with p19, 850 with IL-12R β 1, 223 with IL-23R.
- p19 binds only IL-23R D1 (1234 contacts) and nothing else in the complex.
- IL-12R β 1 never touches p19 (23.5 Å minimum) or IL-23R (11.5 Å): the two receptor subunits do not directly contact.
- IL-23R rotates 18° between the binary complex (5MZV) and the quaternary (6WDQ): the interface is plastic.

Quaternary signaling complex 6WDQ (3.4 Å). Chains: p40 (ochre), p19 (blue), IL-23R (magenta), IL-12R β 1 N-terminal fragment (teal); two N-glycans shown faint.

Both p19 loops dock into a single immunoglobulin domain of IL-23R

p19–IL-23R interface contact map (5MZV): both p19 loops engage the IL-23R N-terminal Ig domain (24–31 primary, 98–118 secondary)



Sources: PDB 5MZV (2.8 Å) – Bloch et al., Immunity 2018, PMID 29287995; 272 p19–IL-23R atom pairs ≤ 5 Å

Residue-residue contact map from PDB 5MZV: 272 p19–IL-23R atom pairs within 5 Å. Cluster blocks: p19 53–60 and 154–165 engage IL-23R 24–31 (primary) and 98–118 (secondary).

A few hotspot residues carry the interaction, and every one is shallow

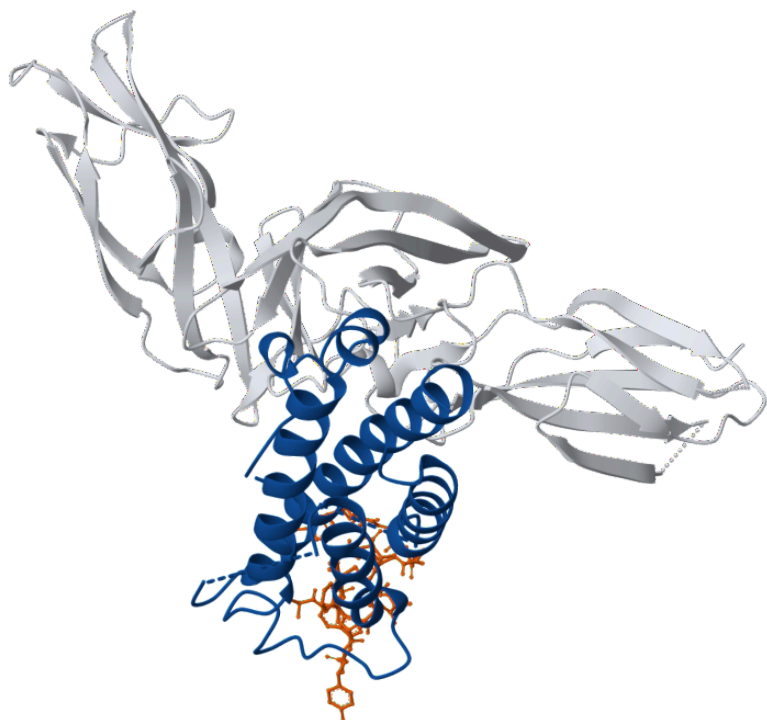


Interface close-up, PDB 6WDQ: p19 53–60 and 154–165 (blue), IL-23R 24–31 and 98–118 (magenta). No glycan on IL-23R in any complex (N29); mouse W157A/L161E abolish binding, mouse K165S/L56E retain it.

P19 RESIDUE	IL-23R PARTNER	DISTANCE / ROLE
W156 (helix D)	G116, D118 (D1)	2.70 / 2.26 Å
K164 (loop 154–165)	G24 (D1 hook)	1.73 Å
R57 (CDR1 53–60)	D1 24–31	charge pattern
L56, L160, L161	D1 24–31	hydrophobic core
E58, E59 (CDR1)	D1 24–31	charge pattern
Cys73–Cys199 (p19–p40)	disulfide	2.03 Å, structural

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This interface is the textbook case of a hard protein–protein interaction for small molecules



PREFORMED POCKET

none

apo IL-23 (8CR8, 2.0 Å) has no pocket to fill

BURIED SURFACE (BSA)

 $\approx 900 \text{ Å}^2$

p19–IL-23R D1 interface (Bloch 2018)

CONTACT RESIDUES

51

weak contacts spread over a flat patch

AFFINITY TO BEAT (KD)

**10–44
nM**

IL-23:IL-23R binding, method-dependent

SMALL-MOLECULE PDBS

0

exhaustive RCSB + PubMed search

BEST FRAGMENT

180 μM 180 μM –1.95 mM, Ferring 2025 HTS (high-throughput screen)

Peptide 23-446 opens a cryptic pocket on p19 (PDB 8UUI, 2.43 Å). Janssen HTS hits act at pathway level, not direct binding; the pocket is not the IL-23R interface.

Five entry points to the interface, ranked by the evidence

LOOP-CLUSTER TARGETING

Aim at the p19 loops

Aim at p19 loops 53–60 and 154–165 on D1 24–31; the K164/R57 to D118/E111/G24 charge pattern is the template.

SELECTIVITY

p19-only by geometry

p19 never contacts IL-12R β 1 (23.5 Å); conformation-specific binding spares IL-12, which shares the p40 subunit.

INDUCED POCKET

Exploit the cryptic pocket

Peptide 23-446 (8UUI) opens a cryptic pocket; peptides 14/15 reach KD 16/20 nM; the 180 μ M fragment gap is headroom.

RECEPTOR SIDE

IL-23R D1 template

The 9MFG VHH epitope (KD 128 pM) overlaps the p19 site; N29 glycosylation and human cross-reactivity unshown.

BIVALENT

p40-assisted anchoring

4GRW nanobodies engage p19 and p40 at once; exploratory, and p40 is shared with IL-12.

CONTEXT

Alternative nodes

Oral TYK2/JAK is a proven route with class-safety baggage; shed IL-23R is a biomarker candidate.

ASSAY LADDER

(i) direct binding: SPR/FP/ITC (binding assays); (ii) NanoBRET competition with fluorescent IL-23 (GSK format); (iii) cell-based signaling with IFN γ counter-screen (Janssen format); (iv) in vivo: no published small-molecule data.

The bar to clear: mAb-level clearance, delivered orally

Week-16 skin clearance by agent; PASI = Psoriasis Area and Severity Index (0–72).

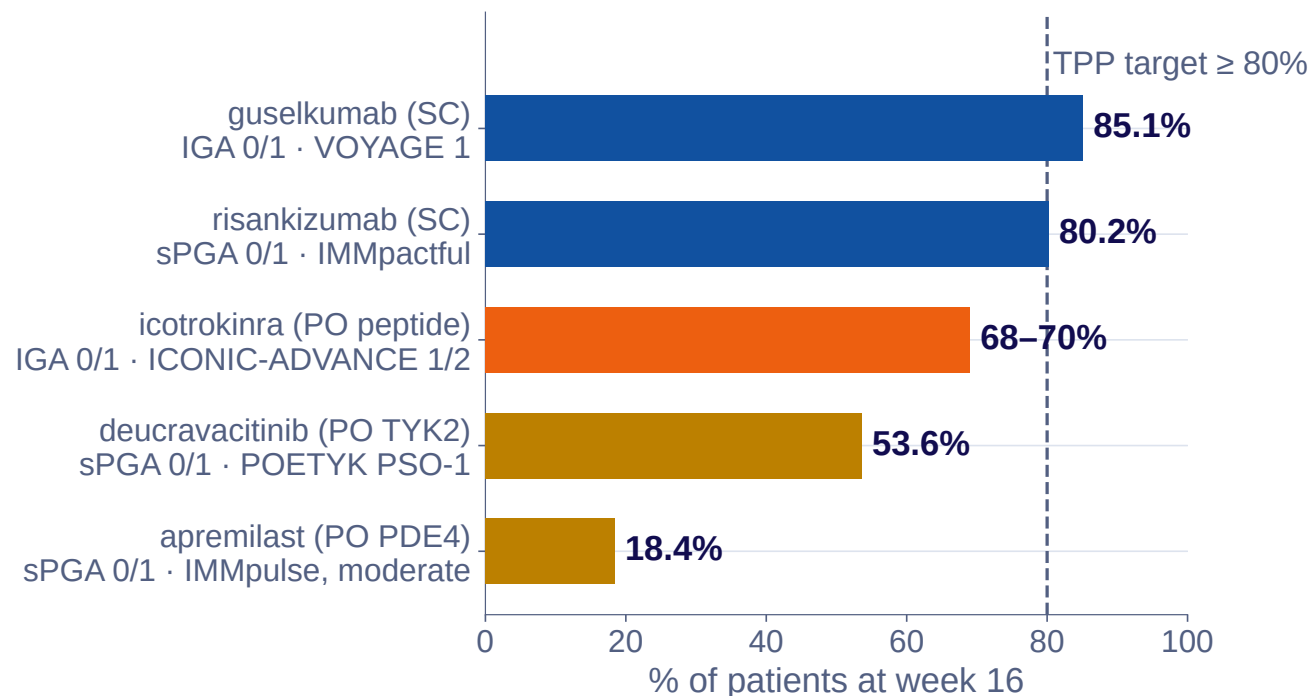
AGENT	CLASS & ROUTE	STATUS	WEEK-16 CLEARANCE	SELECTIVITY / SAFETY SIGNAL
Guselkumab	mAb, SC (subcutaneous) q8w	Approved 2017	IGA 0/1 85.1% (VOYAGE 1)	SI (serious infections) 1.0 per 100 patient-years (PY); no IBD signal
Risankizumab	mAb, SC q12w	Approved 2019; also ulcerative colitis and Crohn's disease	sPGA 0/1 80.2% (IMMpactful)	t _{1/2} ~28 d; no exposure–safety link
Tildrakizumab	mAb, SC q12w	Approved 2018	wk12 PASI-75 62–66% (reSURFACE 1)	ADA 6.5–8.2% (anti-drug antibodies)
Deucravacitinib	oral TYK2 inhibitor	Approved	sPGA 0/1 53.6% (POETIK PSO-1); 39.7% (IMMpactful)	4-yr MACE 0.32/100 PY (major adverse cardiovascular events)
Icotrokinra	oral macrocyclic peptide	Phase 3 results 2025	IGA 0/1 68–70% (ICONIC-ADVANCE)	KD 7.1 pM; IFNγ IC₅₀ 18.4 pM; no IL-12 impact
Apremilast	oral PDE4 inhibitor	Approved	sPGA 0/1 18.4% (IMMpulse, moderate)	PDE4 class; wk16 PASI-90 5.1%

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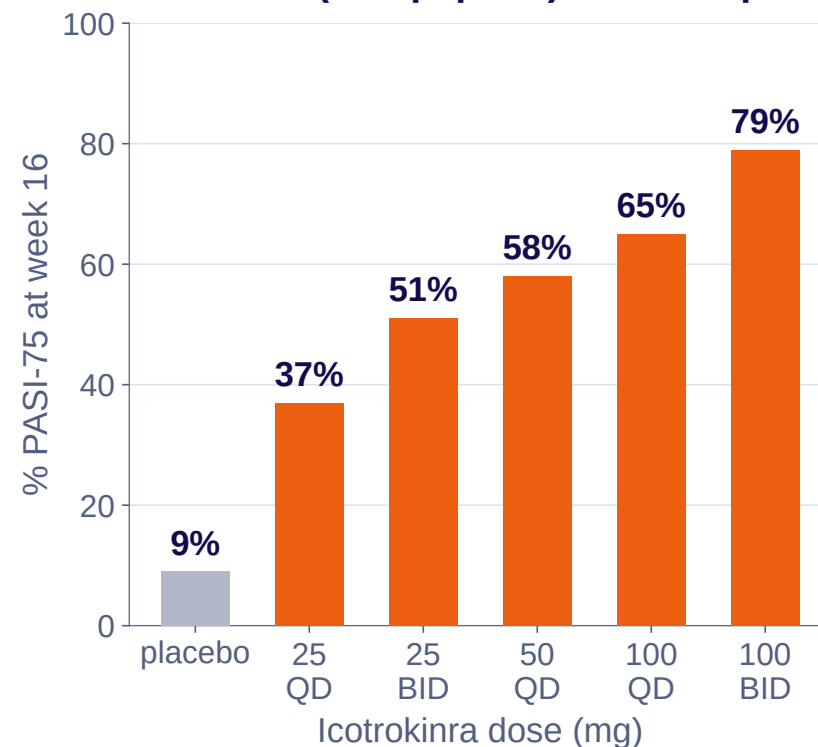
The best oral option sits 12 points below the mAb bar

The best oral option is a peptide (icotrokinra 68–70%); oral small molecules sit at 18–54%.

Week-16 skin clearance by modality



Icotrokinra (oral peptide) dose–response



Sources: VOYAGE 1 · IMMpactful · ICONIC-ADVANCE 1/2 · POETKY PSO-1 · IMMpulse (moderate disease)

Week-16 skin clearance by modality, endpoint per bar (left); icotrokinra FRONTIER-1 PASI-75 dose–response (right). PO = oral · QD = once daily · BID = twice daily. Sources in chart.

The IL-23 class is clean; the JAK class carries the baggage an oral agent must not inherit

MONOCLONAL-ANTIBODY CLASS

Anti-IL-23 / anti-IL-23R: clean

Guselkumab 11-trial pooled safety (4,399 patients): serious infections (SI) 1.0/100 PY, no new-onset IBD signal, no suicidality warning, background-level candidiasis, anti-drug antibodies (ADA) 3.3–8.2%.

Pooled SI	1.0/100 PY
IBD signal	none
ADA rate	3.3–8.2%



JAK CLASS

Oral JAK/TYK2: inherited warnings

ORAL Surveillance (NEJM 2022): tofacitinib raised major adverse cardiovascular events (MACE, hazard ratio HR 1.33) and cancer (HR 1.48) vs TNF blockers in RA patients ≥50 y, with more shingles (HZ). Deucravacitinib's 4-yr MACE of 0.32/100 PY is the reassurance within the class.

MACE HR	1.33
Cancer HR	1.48
Deucravacitinib 4-yr MACE	0.32/100 PY

DESIGN PRINCIPLE

The node is a receptor, not the kinome, so selectivity is structural. The protective IL23R R381Q variant argues the axis is tolerable even when blocked.

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The target product profile: oral once-daily, mAb-level response by week 16

Three preclinical gates: week-16 sPGA 0/1 $\geq$ 80%, oral once-daily (QD), $t_{1/2}$ 12–24 h.

PARAMETER	PRECLINICAL TARGET	BENCHMARK (EVIDENCE)
Efficacy (wk 16)	sPGA 0/1 $\geq$ 80%; PASI-90 $\geq$ 70%; PASI-100 $\geq$ 45–50%	mAb bar 80.2–85.1%; must beat icotrokinra 68–70%
Onset	$\geq$ 30–40% sPGA 0/1 by week 4	IL-23 mAbs reach 50% PASI-90 in ~9–10 wk (IL-17 inhibitors 6–8 wk; NMA 45 trials, PMID 40676461)
Durability	wk 52 IGA 0 $\geq$ 40%; stretch PASI-100 $\geq$ 50%	icotrokinra wk52 IGA0 43% (FRONTIER-2, PMID 39549848); risankizumab wk256 PASI-90 85.1% (LIMMitless, PMID 37553030)
Potency	human Th17 IC50 $\leq$ low nM; binding KD $\leq$ ~44 nM	IL-23:IL-23R KD 10–44 nM; icotrokinra KD 7.1 pM (peptide)
Selectivity	no IL-12 impact; $\geq$ 10–30 $\times$ vs IFN γ ; no IL-22	icotrokinra: no IL-12 impact; IFN γ IC50 18.4 pM
PD (pharmacodynamics)	serum IL-17A/17F/22 reduced wk 1–4, sustained	VOYAGE 1 substudy; ECLIPSE
Safety & PK (pharmacokinetics)	SI $\leq$ ~1.5/100 PY; no IBD; no JAK-class MACE/cancer/HZ; QD; no CYP DDI (drug–drug interaction)	guselkumab pooled SI 1.0; ORAL Surveillance; risankizumab PK

The nonclinical package de-risks two unknowns: species translation and oral exposure

1

Species cross-reactivity

Panel across species plus humanized and transgenic mice; human–mouse IL-23R cross-reactivity is unverified.

2

Human ex vivo potency

PBMC/Th17 potency with IL-12, IFN γ and IL-22 counter-screens.

3

Rat proof of concept

IL-23 skin model for efficacy; rat TNBS colitis to show no IBD signal.

4

Toxicology

28- and 90-day rodent and dog: hERG (cardiac channel), DILI (liver injury), genotox, immunotox.

5

Margins

NOAEL (no observed adverse effect level) margin $\geq 10\times$; exposure–efficacy and exposure–safety mapping.

THE TWO UNKNOWNs THIS PACKAGE MUST CLOSE

(a) Human–mouse IL-23R cross-reactivity is unverified and species interfaces differ (Esch 2020): human readouts are mandatory. (b) Oral absorption at therapeutic exposure is unproven in this class: no published in-vivo small-molecule data; icotrokinra's peptide did it at 100 mg twice daily (BID).

Four structure-driven plays for the p19–IL-23R interface

HOTSPOT MIMICRY

Mimic the W156 cluster

W156 → G116/D118 at 2.7/2.3 Å; mouse W157A abolishes binding. Design a planar aromatic with the R57/K164 charge pattern.

INDUCED POCKET

Exploit the cryptic pocket

The 8UUI cryptic pocket. Fragment KD 180 μM versus peptide 28 nM is the optimization headroom.

RECEPTOR SIDE

Anchor the D1 loops

The G24–N29 hook plus 98–118; the 9MFG VHH template (KD 128 pM, 8 H-bonds) as guide.

GEOMETRY

Use p19-only geometry

IL-12Rβ1 never contacts p19 (23.5 Å): conformation-specific binding spares IL-12.

CAVEATS

The Cys73–Cys199 disulfide is a structural fact, not a covalent target, and no validated covalent warhead exists: covalent remains unexplored, high-risk white space. N29 glycosylation is unknown: verify by glyco-engineering before committing receptor-side leads. Human readouts are mandatory (Esch 2020).

The honest risks, and the white space that makes the program worthwhile

RISKS

Five hard ones

- A flat 900 Å² interface is slow, hard chemistry.
- Species translation is unverified; human–mouse interfaces differ.
- No in-vivo small-molecule data anywhere: oral exposure is an assumption.
- A strong peptide competitor (icotrokinra, phase 3) is close.
- The mAb bar keeps moving (durability data through week 256).

WHITE SPACE

Five open doors

- 0 small-molecule PDBs: first-mover structural data goes here.
- No covalent options exist on this target.
- The induced pocket is a novel mechanism.
- p19-only selectivity is a class-first property.
- No in-vivo small-molecule data: an unclaimed proof of concept.

Every number in this deck traces to a public primary source

WHAT IT SUPPORTS	STRUCTURE / TRIAL	CITATION
Quaternary complex (two-tier interface)	6WDQ (3.4 Å)	Glassman et al., Cell 2021 (PMID 33606986)
First complex + affinity	5MZV (2.8 Å)	Bloch et al., Immunity 2018 (PMID 29287995)
Apo IL-23, no preformed pocket	8CR8 (2.0 Å)	Bloch et al., NSMB 2024 (PMID 38287195)
Induced pocket + peptide SAR	8UUI (2.43 Å)	Lecomte et al., ACS Chem Biol 2025 (PMID 40560919)
Receptor-side VHH epitope	9MFG (2.85 Å), KD 128 pM	Ota et al., PNAS 2025 (PMID 40434646)
Efficacy & onset benchmarks	VOYAGE 1 · IMMpactful · ICONIC-ADVANCE · NMA onset (45 trials)	PMID 28057360 · 42143644 · 40976249 · 40676461
Durability & safety anchors	guselkumab pooled · ORAL Surveillance · FRONTIER-2 · LIMMitless	PMID 37906417 · 35081280 · 39549848 · 37553030

VERIFICATION NOTES

Structures verified against RCSB Protein Data Bank coordinates and PubMed full text; trial numbers from primary papers. ClinicalTrials.gov and patent databases were inaccessible (HTTP 403), so IND and patent claims are outside scope. Known-false claims (the 'Schreiber small-molecule PDB') are excluded.

The interface is hard. The data to attack it are public.

Proceed with a p19-centric fragment-to-lead campaign, run the receptor-side VHH epitope as a parallel template, and gate candidate selection against the preclinical TPP: week-16 sPGA 0/1 $\geq$ 80%, oral once-daily, mAb-class safety.